

Chapter 2: Continuous Random Variables

UE25MA141B
Engineering Mathematics-II

1 Continuous Random Variables

1.1 Introduction

Thus far, we have studied Discrete random variables, that is, random variables whose support is either finite or countably infinite. However, there also exist random variables whose support is uncountable. For example, the time that a bus arrives at a bus stop. The lifetime of a radioactive material etc. Let X be such a random variable. We say that X is a continuous random variable if there exists a non negative function f such that for all real numbers $x \in (-\infty, +\infty)$, having the property that, for any set B of real numbers,

$$P(x \in B) = \int_{x \in B} f(x) dx \quad (1)$$

The function f is called the density function of the random variable X . See figure below

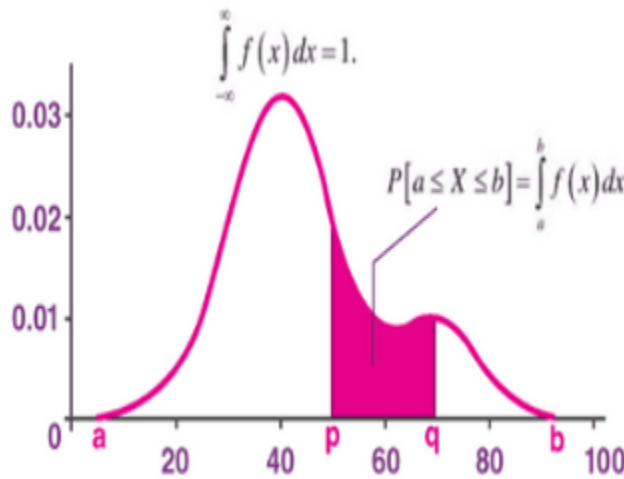


Figure 1: Probability density function of a continuous random variable

This means that the probability that a continuous random variable X assumes values in an interval $[p, q]$ is equal to the area of the region bounded by the density function $y = f(x)$, the X -axis and the lines $x = p$ and $x = q$.

$$P(p \leq X \leq q) = \int_p^q f(x) dx \quad (2)$$

Remark 1. For all numbers x , the density function $f(x) \geq 0$, so that the graph of $y = f(x)$ never drops below the x -axis.

Remark 2. The area under the density function $y = f(x)$ and above the x -axis is equal to 1.

Remark 3. Because area of a line segment is zero, the probability that a continuous random variable takes on exactly one value is zero.

Remark 4. Thus whether or not the end points of an interval are included makes no difference concerning the probability of the interval. i.e

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) \quad (3)$$

Examples

Example 1. The random variable X has a uniform distribution on $[0, 1]$: The density function $f(x) = 1$ if $0 \leq x \leq 1$ and $f(x) = 0$ for all other values of x . See figure below

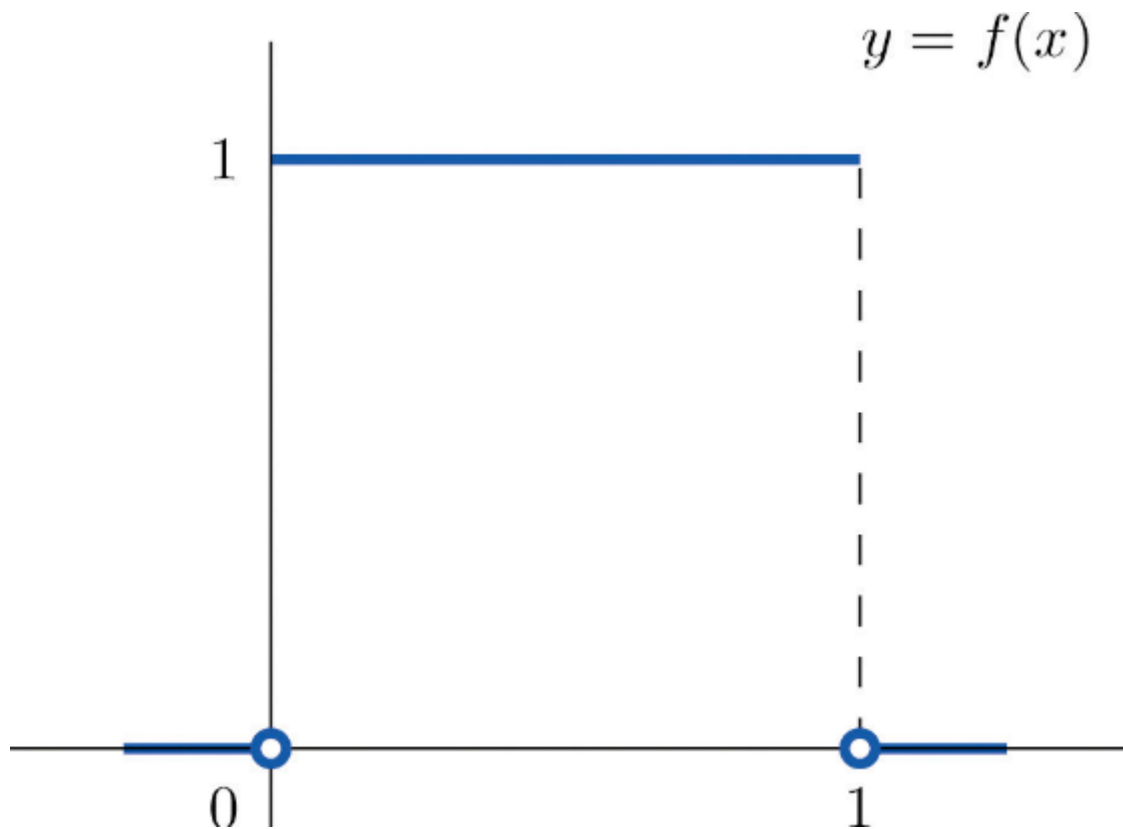


Figure 2: Uniform distribution on $[0, 1]$

- a) Find probability that $X > 0.75$.
- b) Find $P(X \leq 0.2)$.

c) Find $P(0.4 < X < 0.7)$.

Solution:

a) By definition $P(X > 0.75)$ equals the area under the uniform density function $f(x)$ between 0.75 and 1. See Fig 3 below.

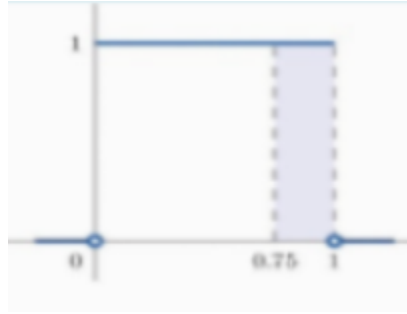


Figure 3: Shaded area corresponds to $P(X \geq 0.75)$

Thus, $P(X \geq 0.75) = \text{area of the shaded rectangle} = 0.75 \times 1 = 0.75$

b) $P(X \leq 0.2) = \text{area under the uniform density function between 0 and 0.2}$. See Fig 4

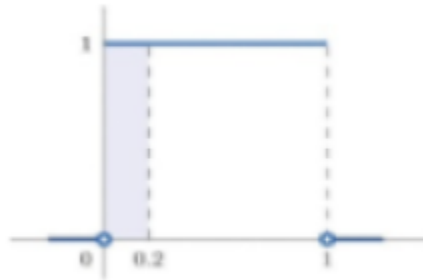


Figure 4: Shaded area corresponds to $P(X \leq 0.2)$

Thus, $P(X \leq 0.2) = 0.2 \times 1 = 0.2$.

c) $P(0.4 < X < 0.7) = \text{area under the uniform density function from 0.4 to 0.2 having height 1 unit}$. This gives $P(0.4 < X < 0.7) = 0.2 \times 1 = 0.2$. See Fig 5 below.

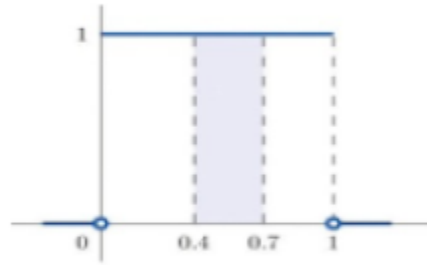


Figure 5: Shaded portion corresponds to $P(0.4 < X < 0.7)$

Example 2. The amount of time, in hours, before a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} \lambda e^{-x/100} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

What is the probability that

- a) a computer will function between 50 and 150 hours before breaking down;
- b) it will function less than 100 hours

Solution:

a) Since $\int_{-\infty}^{\infty} f(x) dx = 1$, we have, $\lambda \int_0^{\infty} e^{-x/100} dx = 1 \implies \lambda = \frac{1}{100}$.

Hence the probability that a computer will function between 50 and 150 hours before breaking down is:

$$P(50 < X < 150) = \frac{1}{100} \int_{50}^{150} e^{-x/100} dx = -e^{-x/100} \Big|_{50}^{150} = e^{-1/2} - e^{-3/2} \approx 0.384$$

b) The probability that it will function less than 100 hours = $P(X < 100) = \int_0^{100} f(x) dx = \frac{1}{100} \int_0^{100} e^{-x/100} dx = -e^{-x/100} \Big|_0^{100} = 1 - e^{-1} \approx 0.633$

1.2 Expectation and Variance of Continuous random variables

We defined expected value of a discrete random variable by $E[X] = \sum_x x P[X = x]$. If X is a continuous random variable having density function $f(x)$ then the analogous definition is to define the expected value of X as :

Expectation of a continuous random variable

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx \quad (4)$$

Expectation of a function g of a continuous random variable

If X is a continuous random variable having density function $f(x)$ then the expectation of any real-valued function $g(x)$ of X is defined as:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx \quad (5)$$

This is also known as the *Law of unconscious statistician*

Examples

Example 3. Find $E[X]$ if the density function of X is defined as:

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

$$E[X] = 2 \int_0^1 x dx = 2 \left\{ \frac{x^2}{2} \right\} \Big|_0^1 = 1$$

Example 4. If X is a continuous random variable with distribution function $F_X(x)$ and density function $f_X(x)$, find the density function of the random variable Y defined by $Y = 2X$.

Solution: Given that $F_X(x) = P(X \leq x)$.

Therefore,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(2X \leq y) \\ &= P(X \leq \frac{y}{2}) \\ &= F_X(\frac{y}{2}) \end{aligned}$$

The density function of Y is obtained by differentiation the above result on both sides. Thus:

$$f_Y(y) = \frac{1}{2} f_X\left(\frac{y}{2}\right)$$

Example 5. The density function of X is given by

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[e^X]$ Solution: Let $Y = e^X$. Thus Y is a function of a random variable X . By the law of unconscious statistician, we have

$$E[e^X] = \int_0^1 e^x f_X(x) dx$$

$$\text{where, } f_X(x) = \int_0^1 dx = 1,$$

Therefore,

$$E[e^X] = \int_0^1 e^x dx = e^x \Big|_0^1 = e - 1$$

1.3 Chebyshev's Inequality

Let X be a random variable with mean $E[X] = \mu_X$ and standard deviation σ_X . Then,

$$P(|X - \mu_X| \geq k\sigma_X) \leq \frac{1}{k^2} \quad (6)$$

This means that, the probability that a random variable differs from its mean by k standard deviations or more is never greater than $\frac{1}{k^2}$.

Example 6. The length of a rivet manufactured by a certain process has mean $\mu_X = 50mm$ and standard deviation $\sigma_X = 0.45mm$. What is the largest possible value for the probability that the length of the rivet is outside the interval $(49.1, 50.9)mm$?

Solution: From Chebyshev's Inequality, we have

$$P(X \leq 49.1 \text{ or } X \geq 50.9) = ?$$

$$P(X \leq 49.1 - 50 \text{ or } X \geq 50.9 - 50) = \frac{1}{2^2}$$

$$P(|X - 50| \geq 0.9) =$$

$$P(|X - \mu_X| \geq 2(\sigma_X)) \leq \frac{1}{2^2}$$

1.4 Bernoulli Distribution

Suppose that a trial, or an experiment, whose outcome can be classified as either a success or a failure is performed. If we let $X = 1$ when the outcome is a success and $X = 0$ when the outcome is a failure, then the probability mass function of X is given by

$$\begin{aligned}P(X = 0) &= p(0) = 1 - p, \\P(X = 1) &= p(1) = p\end{aligned}\tag{7}$$

Where p , with $0 \leq p \leq 1$, is the constant probability that the trial is a success. See Figure 6 below: The random variable X is said to be a Bernoulli random variable (after the Swiss mathematician James Bernoulli) if its probability mass function is given by (7) for some $p \in (0, 1)$.

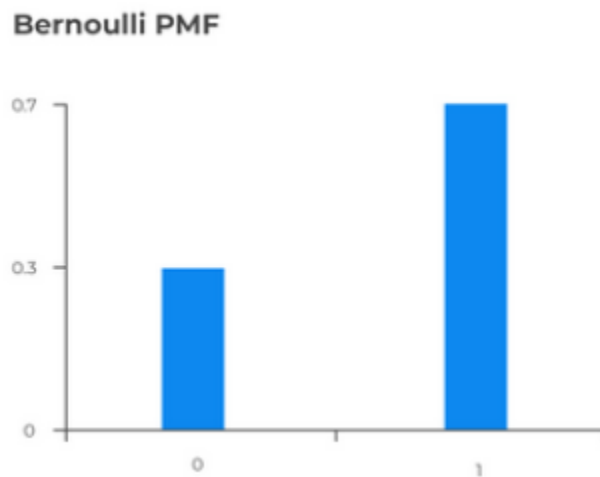


Figure 6: Pmf of Bern(0.7) random variable

Remark 5. Bernoulli distribution often serves as a starting point for more complex distributions. For example: Bernoulli process lays a foundation for binomial distribution, geometric distribution, negative binomial distribution, all of which play a crucial role in advanced probability.

Remark 6. Bernoulli distribution is a discrete probability distribution.

Remark 7. Bernoulli distribution describes the probability of achieving a success or a failure.

Remark 8. A Bernoulli trial is an even that has only 2 possible outcomes (success or failure). For example: A coin will land heads or tails.

Remark 9. In experiments and clinical trials, the Bernoulli distribution sometimes is used to model a single individual experiencing an event like death, disease or disease exposure. The model is an excellent indicator of the probability a person has the event in question.

1.5 Cumulative distribution function of a Bernoulli random variable

The CDF of a Bernoulli random variable is given by

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases} \quad (8)$$

The cdf may be represented graphically as in Figure 7:

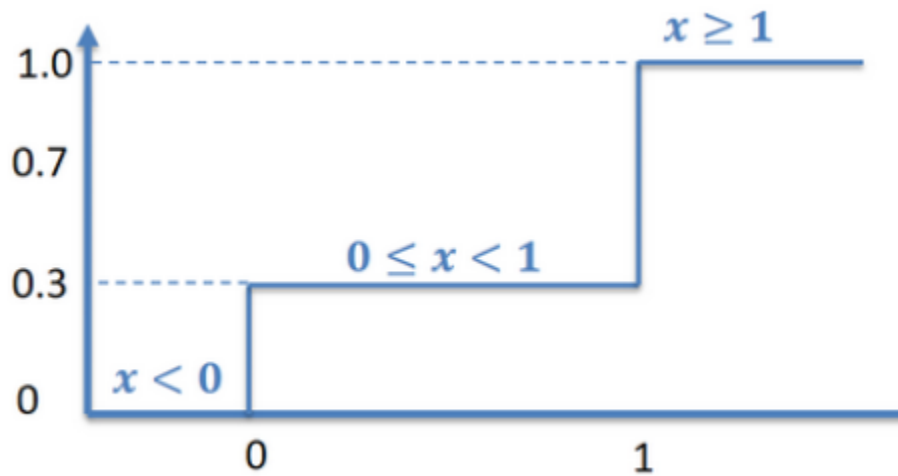


Figure 7: CDF of a Bern(0.7) random variable

1.6 Conditions under which a Bernoulli distribution is defined:

1. Only two possible outcomes are possible.
2. Each of the outcomes has a fixed probability p of occurring.
3. Trials are entirely independent of each other.

Examples

Example 7. An archer can hit the target with probability 0.6. What is the probability that he misses the target ?

Solution: Probability (Archer hits the target) = $P(\text{success}) = 0.6$.

Hence, probability(Archer misses the target) = $P(\text{ failure}) = 1 - 0.6 = 0.4$

Example 8. If a Bernoulli distribution has a parameter 0.45. Then find its expectation.

Solution: Let $X \sim \text{Bern}(p)$. Then $E[X] = p$.

Hence if $X \sim \text{Bern}(0.45)$. Then $E[X] = 0.45$.

1.7 Mean and Variance of a Bernoulli distribution

Recall that the general formula for Mean or Expectation of a Discrete random variable is defined as :

$$\sum_{n=0}^{\infty} x_n P[X = x_n] \quad (9)$$

That is, the expectation of a discrete random variable is the sum of the products of all outcomes and their corresponding probabilities. Thus, for a Bernoulli random variable, we have

$$\begin{aligned} E[X] &= 0 \times P[X = 0] + 1 \times P[X = 1] \\ &= 0 \times (1 - p) + 1 \times (p) \\ &= p \end{aligned} \quad (10)$$

The variance of a discrete random variable is, by definition,

$$\text{Var}(X) = E[X^2] - \{E[X]\}^2 \quad (11)$$

Where $E[X^2] = \sum_0^1 x^2 P(X = x) = 1 \cdot p = p$. Let $q = 1 - p$. This gives,

$$\text{Var}(X) = p - p^2 = p(1 - p) = pq \quad (12)$$

Remark 10. If the experiment is a deterministic experiment with $p = 0$, i.e the outcome is impossible. Hence the variance must be zero as can be verified from the expression for variance.

Remark 11. If the experiment is a deterministic experiment with outcome $p = 1$, then the outcome is certain. Hence the variance is again 0.

Remark 12. Sum of independent Bernoulli random variables is a Binomial random variable. This will be proved and discussed in the lecture entitled "Binomial distributions"

1.8 Higher Moments of the Bernoulli random variable

The r^{th} moment of a Bernoulli random variable is given by

$$\begin{aligned} E[X^r] &= \sum_0^1 x^r P(X = x) \\ &= 0 \times (1 - p) + 1 \times p \\ &= p = E[X] \end{aligned} \tag{13}$$

Example 9. Let X be a Bernoulli random variable with parameter $P = \frac{1}{2}$. Find the tenth moment of X .

Solution: By definition, $E[X^r] = E[X]$ for a Bernoulli random variable. Hence the tenth moment of X is $E[X^{10}] = E[X] = p = \frac{1}{2}$.

2 Miscellaneous Examples

1. Consider the event of tossing an unbiased coin. a) What is the probability of getting heads?
b) What is the probability of getting tails?

Solution: Sample space : $S = \{H, T\}$. Therefore, $P(H) = \frac{1}{2}$ and $P(T) = \frac{1}{2}$.

2. Consider an unfair coin with probability of success (p) or head is 0.8 and the probability of failure (q) or tail is 0.2. What is the expectation of the random variable that counts the number of heads.

Solution: $E[X] = p = 0.8$

3. The prevalence of a certain disease in a given population is 10 percent. What is the probability that a random person from this population is healthy?

Solution: If we randomly select a person from this population, there are only two possibilities. A person is either healthy or diseased. It is given that the probability that a person is diseased is $p = 10$ percent. Thus,

$$P(\text{person is healthy}) = 1 - p = 1 - 0.1 = 0.9$$
